

Homework Classical Physics, Fall 2023

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Problem Set 03.4 (L05 & L06)

Due on Thursday, 28. September 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - Friction and acceleration (4 points)

A block of mass $m = 0.45 \text{ kg}$ lies on a wedge of $m = 2.0 \text{ kg}$ which has an angle of its slope of 35° . The wedge slides frictionless on a horizontal plane due to a horizontal force F . The static frictional coefficient between block and wedge is $\mu_{sf} = 0.70$. What is the maximum and minimum value of the force F in between which the block does not slide (up or down) on the wedge?

